

Robotic Navigation Using Large Language Models

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ABSTRACT

This study investigates the potential of large language models (LLMs), specifically GPT-4 (4o and 4Turbo), to assist in robotic navigation tasks within a simulated maze. Despite LLMs being fundamentally trained as next-token predictors on internet-scale data, they demonstrate emergent reasoning and perception capabilities that may be applicable to spatial navigation challenges. Our approach combines visual data from cameras and point clouds projected onto 2D surfaces with natural language instructions and contextual information, prompting the models to generate environmental summaries and directional guidance toward specified goals. Results reveal both promising capabilities and limitations. Notably, our findings indicate that models with built-in multi-modal capabilities significantly outperform those where such capabilities were added post-training. This research provides insights into the feasibility of LLMs as components in robotic navigation systems and identifies promising directions for future work in this domain.

INTRODUCTION

Autonomous navigation represents a cornerstone capability in modern robotics, enabling machines to traverse complex environments without constant human supervision. Traditionally, this domain has relied on specialized algorithms and control systems meticulously engineered for specific navigation tasks. However, the robotics landscape needs a paradigm shift as current state-of-the-art navigation systems remain constrained by their lack of emergent capabilities.

The fundamental challenge in autonomous navigation lies in creating systems that can process high-level instructions and translate them into precise navigational actions—a capability humans perform effortlessly. While conventional approaches typically require extensive pre-training on specific environments or rely on hard-coded rules, there remains a critical gap in developing systems that can navigate novel environments through natural language interaction. The aspiration, as articulated in vision-language-action research, is to create machines that respond to verbal commands and accomplish physical tasks in a manner that is both predictable and safe.



Figure 1: The Jackal robot platform used in our navigation experiments

Unlike autonomous vehicles that leverage extensive pre-mapping and environmental data, our focus is on navigation scenarios analogous to human drivers encountering unfamiliar terrain for the first time. This zero-shot navigation capability, where the system is not fine-tuned for specific environments, represents a significantly more challenging but potentially more versatile approach. The LLM in this context serves as an adaptive reasoning engine rather than a repository of pre-trained environmental knowledge. To evaluate this capability, we conducted experiments in controlled environments including the maze configuration shown in Figure 2, which presents complex decision points and varied navigational challenges.

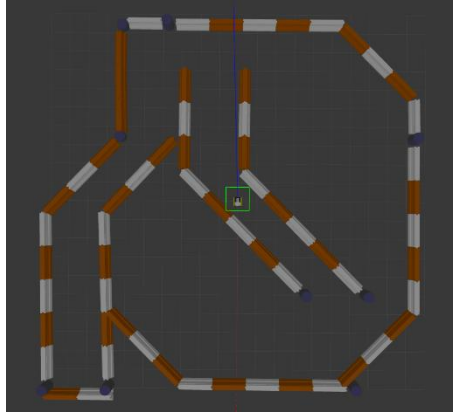


Figure 2: Experimental maze environment used to evaluate navigation capabilities

This research aims to investigate how multimodal LLMs, particularly models in the GPT-4 family, can interpret visual data in real-time and generate appropriate navigational instructions. Our preliminary investigations reveal an intriguing finding: despite expectations that GPT-4 Turbo would outperform due to a higher number of parameters, GPT-4o demonstrated superior navigational reasoning. We attribute this to GPT-4o's natively integrated multimodal architecture, designed from inception to seamlessly process and synthesize visual, textual, and audio inputs. In contrast, GPT-4 Turbo's multimodal capabilities appear to function more as adjunct features rather than core components of its architecture.

The significance of this work extends beyond academic interest. By developing a modular vision-language-action framework compatible with various LLMs, we create a versatile platform for robotic navigation. While purpose-built navigation systems excel in structured, familiar environments, LLM-powered navigation potentially offers unprecedented adaptability in novel scenarios through its capacity for reasoning and generalization. This hybrid approach—combining the reliability of dedicated navigational systems with the flexibility of LLM reasoning—could dramatically enhance robotic autonomy across diverse applications from search and rescue to extraterrestrial exploration. Our "shoestring"

implementation demonstrates that even without custom training or expensive specialized hardware, existing LLMs can be leveraged to create functional vision-language-action systems—potentially democratizing access to advanced navigational capabilities for robotics researchers and developers worldwide.

EXPERIMENT

Design

To evaluate our navigation system, we developed a simulated testing environment using Robot Operating System (ROS) integrated with Gazebo simulator running on an older Ubuntu distribution to ensure compatibility with essential ROS packages. This configuration required several custom hardware patches to accommodate the specialized sensing requirements of our experiments.

The simulation featured a moderate-complexity maze environment with interconnected corridors, multiple decision junctions, and dead ends to challenge the navigation capabilities of our system (Figure 2). We designed two distinct navigation objectives with increasing difficulty levels: first, requiring the robot to contact a grey wall and then locate and touch an orange wall after executing a 180° turn — a relatively less challenging task due to the abundance of grey and orange walls throughout the environment; second, directing the robot to locate and approach a blue pillar—a deliberately more challenging task due to the limited presence of blue pillars within the environment. The experimental design intentionally restricted environmental knowledge provided to the LLM to necessitate genuine navigational reasoning rather than relying on pre-mapped routes. The Jackal robot platform was equipped with two primary sensing modalities: a front-facing RGB camera providing visual feed of the environment (Figure 3), and a LIDAR sensor generating a 2-D flattened point cloud data representing spatial geometry of the surroundings (Figure 4). This sensor configuration allowed us to test the LLM's ability to integrate visual recognition

with spatial awareness, a fundamental requirement for autonomous navigation in unfamiliar environments.

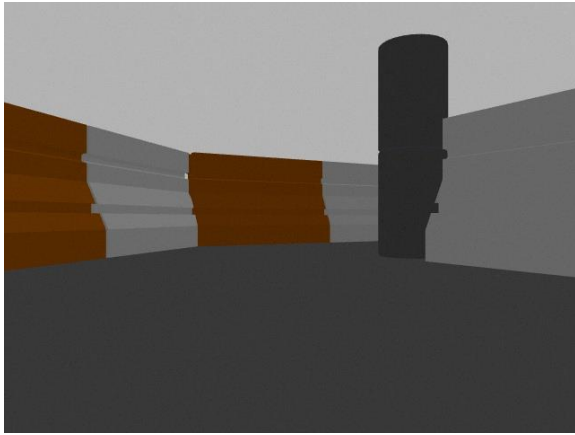


Figure 3. RGB Front Cam Image

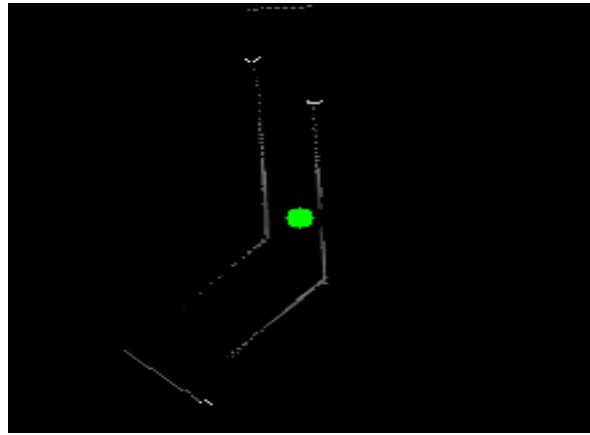


Figure 4. Point Cloud Image

Method

We conducted a comprehensive evaluation comparing two distinct methods across four experimental quadrants: GPT-4o with the grey-orange wall goal, GPT-4o with the blue pillar goal, GPT-4 Turbo with the grey-orange wall goal, and GPT-4 Turbo with the blue pillar goal. Each method represented a unique combination of model architecture and navigational objectives, allowing us to isolate the effects of both factors on performance. The primary success metric was navigation efficiency, measured by the number of interaction rounds required to successfully achieve the designated goal, with a maximum cutoff of 20 rounds establishing a "Did Not Finish" (DNF) outcome. To ensure methodological rigor and statistical validity, we conducted 10 experimental rounds for each quadrant, with each round featuring a different starting position and orientation within the maze environment. Importantly, these starting configurations were kept identical across all four quadrants, enabling direct performance comparisons while controlling environmental variability. We also tracked a secondary metric of "discards," instances where the LLM returned JSON with extraneous or incomplete data that did not conform to our required format specifications,

necessitating rejection of the response and requiring regeneration. This metric provided additional insight into each model's precision and consistency in producing structured navigational commands. This experimental design allowed us to systematically evaluate both the impact of model architecture (comparing GPT-4o's native multimodal integration against GPT-4 Turbo's capabilities) and task complexity (contrasting the more abundant grey-orange wall navigation against the scarcer blue pillar target) on zero-shot navigational reasoning performance.

Procedure

Our implementation utilized Python as the core orchestration framework to scaffold the navigation system and integrate its diverse components. The main control loop began by capturing two crucial data sources: the RGB image from the front-facing camera and the LIDAR point cloud data, which was subsequently processed and flattened into a top-down 2D representation for improved spatial interpretation. These visual inputs, along with the navigational prompt, were then submitted to the selected LLM (either GPT-4o or GPT-4 Turbo) through their respective APIs. The LLM processed this multimodal input and returned a structured JSON response containing three key elements: a situational summary describing the robot's current environment and position, specific navigational directions (decomposed into discrete turn and move commands with appropriate parameters), and a 'done' flag indicating whether the goal had been achieved. Our Python controller then translated these commands into the appropriate ROS action messages and submitted them to the robot's movement controllers for execution. If the 'done' flag remained false, the system would initiate another iteration, resubmitting the original prompt along with the LLM's previous situational summary and newly captured sensor imagery, creating a closed-loop system that continuously refined its navigational decisions based on updated environmental perception and accumulated contextual understanding. Figure 5 displays the flowchart of this procedural

loop, illustrating the sequential data flow and decision points that comprise our vision-language-action navigation system.

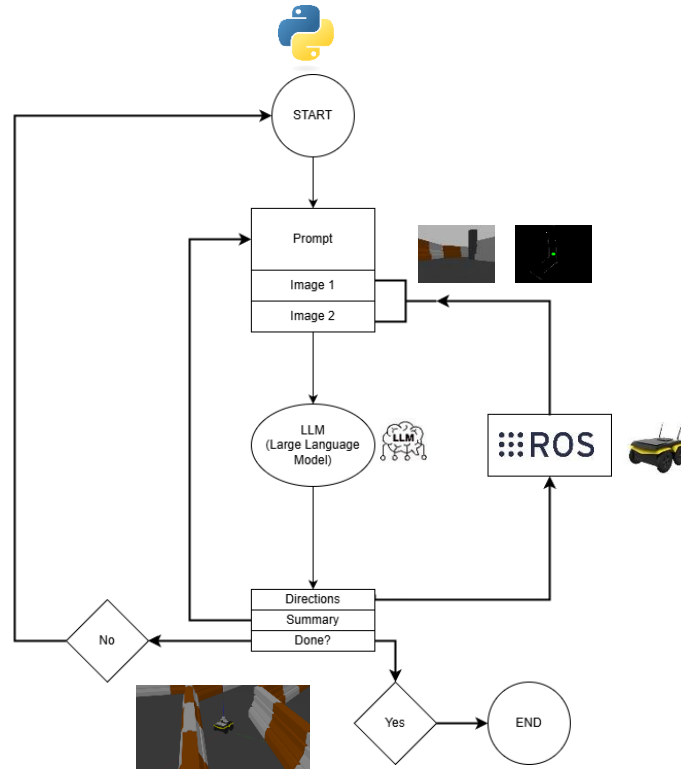


Figure 5. Procedure Loop

Prompt Design

The prompt engineering process for the LLM navigation experiment involved strategic adaptations based on observed model behaviors during test runs. Initially, a recurring issue was identified where the model would execute 180-degree turns unnecessarily. This observation led to prompt modifications that explicitly encouraged the model to mention that a 180-degree turn has been made in the summaries. Table 1 shows the comprehensive prompt design structure, which consists of distinct system and user components. The system section clearly establishes the navigation parameters through multiple critical elements: the goal/target definition, allowed movement constraints, available data specifications, suggested navigation strategy, additional contextual notes, expected

response format, and specific instructions governing response generation. The user component provides environmental perception data, including LiDAR readings and front camera imagery, which serves as the primary input for the model's decision-making process. A notable refinement in the prompt design was the integration of summaries in subsequent prompts, allowing the model to maintain contextual awareness of its previous decisions and environmental understanding, thereby enabling more consistent navigation behavior. This iterative prompt optimization process directly addressed emergent patterns in the model's learning within the navigation environment.

Role	Content
System	- Goal/Target - Movements Allowed - Available Data - Navigation Strategy - Additional Notes - Expected Response Format - Instructions for Response
User	Lidar and Front camera images
AI	Summary

Table 1. Prompt Design

LIMITATIONS

The experimental framework exhibited several notable constraints that impacted performance outcomes. The navigation system suffered from significant perceptual challenges, particularly when the agent encountered close-proximity obstacles. In instances where the robot directly faced a wall within the minimum detection threshold of the point cloud sensors, the system consistently entered a deadlock state, unable to generate viable navigation commands. This limitation stemmed from sensor blind spots in the immediate

vicinity of the robot, creating zones where environmental data was effectively unavailable for decision-making processes. Additionally, the large language model demonstrated a persistent reluctance to utilize negative distance parameters in its movement calculations, likely attributable to insufficient contextual reference for such maneuvers within its training paradigm. The experimental procedure required substantial prompt engineering refinements prior to formal trials to establish appropriate behavioral boundaries and physical constraints of the robotic platform. These adjustments were necessary to prevent commands that exceeded mechanical capabilities or would result in collision events. Furthermore, despite explicit instructions in the prompt design requesting concise situational summaries to maintain contextual awareness across sequential interactions, the model consistently underperformed in generating these synopses, limiting its ability to maintain operational continuity through extended navigation sequences. Complex environmental features, including maze structures with multiple dead ends and areas prone to sensor occlusion, further challenged the system's ability to maintain effective localization and path planning.

RESULTS

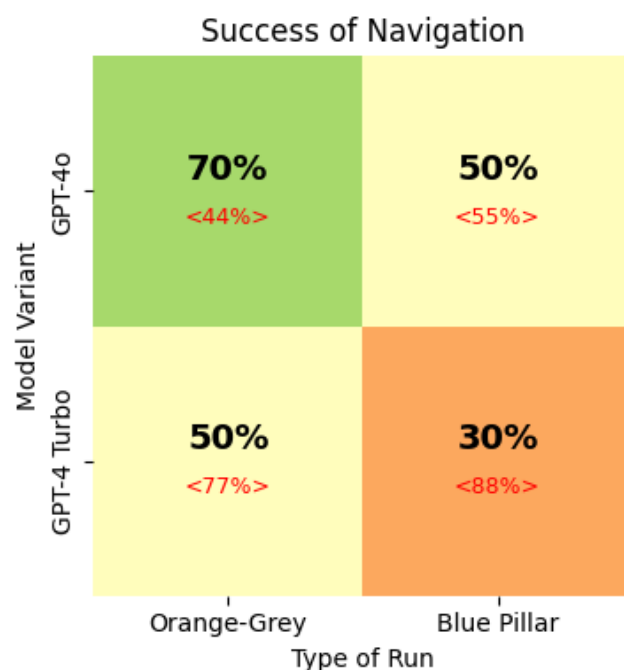


Figure 6. Success/Discard Matrix

Analysis

The comparative performance analysis between language models reveals architectural and cognition differences in navigational capabilities. GPT-4o demonstrated superior task completion capabilities with a markedly higher success rate and fewer discarded attempts than its counterparts, as evidenced in Table 2. This enhanced reliability suggests a more robust interpretation of environmental and navigation parameters. However, this operational consistency came at the cost of increased computational iterations, with GPT-4o requiring significantly more decision-making rounds to achieve successful navigation outcomes.

Conversely, GPT-4 Turbo exhibited contrasting performance characteristics. Despite its distinct architecture, including larger number of parameters that theoretically enable more sophisticated reasoning, GPT-4 Turbo displayed diminished navigational accuracy. The performance matrix presented in Figure 6 quantifies this disparity, with bold numerals representing successful completion percentages across 10 experimental trials for each model, while the bracketed values indicate the proportion of discarded attempts due to unresolvable navigation states or persistent errors.

This inverse relationship between model complexity and navigational precision suggests that parameter count alone does not determine efficacy in embodied decision-making tasks. Instead, the results indicate that specialized architectural elements or training methodologies within GPT-4o may better facilitate the interpretation of spatial data and generation of appropriate motion commands. The higher discard rate observed in GPT-4 Turbo trials further implies potential issues with environmental perception or decision consistency, possibly stemming from overfitting to non-navigational tasks during its training regimen.

These findings highlight the importance of model-specific optimization for robotic navigation applications, where the balance between computational efficiency and task completion reliability remains a critical consideration for deployment in real-world scenarios.

Task	GPT-4o		GPT-4 Turbo	
	Average Number of Rounds	Average Number of Rounds When Successful	Average Number of Rounds	Average Number of Rounds When Successful
Orange-Grey	13.3	12.43	7.2	9
Blue Pillar	13.4	7.2	5	3.67

Table 2. Trial Analysis

Interpretations and Observations

The experimental trials revealed significant insights into the operational dynamics of large language models in robotic navigation contexts. A critical observation emerged regarding the integration of multimodal sensory inputs; the system frequently demonstrated an inability to establish coherent relationships between point cloud data and front camera imagery. This perceptual disconnect resulted in inconsistent environmental interpretations, particularly when point cloud readings indicated navigable pathways that were not properly reconciled with visual data, suggesting fundamental limitations in cross-modal reasoning capabilities.

The architectural distinctions between GPT-4o and GPT-4 Turbo appeared to substantially influence performance characteristics. GPT-4o, with its natively integrated multimodal design architecture, demonstrated superior navigational decision-making compared to GPT-4 Turbo, which incorporated multimodal capabilities through subsequent architectural modifications. This performance differential, while acknowledging the

proprietary and closed-source nature of these models, strongly suggests that foundational integration of multimodal processing yields more effective embodied cognition than retrofitted approaches.

Prolonged operational trials consistently revealed algorithmic tendency toward recursive behavior patterns, with navigation sequences frequently devolving into repetitive action loops after approximately 20 decision cycles, justifying our established experimental cutoff threshold. This behavioral entrenchment indicates limitations in maintaining contextual awareness across extended operational sequences, despite prompt engineering efforts to enhance temporal reasoning.

GPT-4 Turbo exhibited a notably higher frequency of discarded trials, particularly in expansive environmental configurations. This pattern suggests specific deficiencies in processing open spatial contexts, where feature sparsity may have compromised the model's ability to establish reliable navigational references. Contrary to anticipated outcomes regarding learning capabilities, longitudinal performance analysis indicated deterioration rather than improvement over sequential trials, as the systems consistently reproduced suboptimal navigation strategies without apparent adaptation or refinement.

Spatial reasoning limitations manifested most prominently in open-field navigation scenarios, where the models demonstrated impaired distance estimation capabilities. This cognitive constraint resulted in consistently conservative movement parameters when traversing expansive areas, with the system defaulting to smaller displacement increments despite the absence of proximal obstacles. This behavior pattern reveals fundamental challenges in conceptualizing relative spatial relationships within unstructured environments, particularly when interpreting depth information from visual and point cloud data simultaneously.

CONCLUSION

This study demonstrates that large language models can effectively serve as navigational reasoning engines for robotic systems despite being fundamentally text generators. Our comparative analysis between GPT-4o and GPT-4 Turbo revealed that architectural design significantly impacts navigation performance, with GPT-4o's natively integrated multimodal capabilities exhibiting superior environmental interpretation and decision-making consistency compared to GPT-4 Turbo's retrofitted approach. This finding challenges conventional assumptions that parameter count alone determines model efficacy in embodied reasoning tasks, while also highlighting critical limitations in cross-modal integration between point cloud data and visual imagery, recursive behavioral patterns, and diminishing performance over extended operational sequences.

The observed performance differences between model architectures underscore the importance of specifically designed multimodal integration for effective embodied cognition, suggesting that future developments should prioritize foundational multimodal capabilities rather than post-training adaptations. While current LLM-based navigation systems may not yet match the precision of purpose-built solutions, they demonstrate remarkable adaptability and zero-shot reasoning capabilities that conventional navigation algorithms typically lack, establishing a promising foundation for hybrid systems that leverage complementary strengths of traditional algorithmic approaches and the emergent reasoning capabilities of large language models. As these architectures continue to evolve, their potential applications extend beyond navigation to broader domains of physical reasoning and environmental interaction, potentially transforming how autonomous systems interpret and engage with the world.

FUTURE WORK

Building upon the insights gained from this study, several promising research directions warrant further investigation. We propose evaluating advanced reasoning models such as o1 and DeepSeek-R1 on specialized navigation tasks designed to challenge their spatial cognition capabilities and temporal reasoning faculties. These models' enhanced logical processing architectures may offer superior performance in complex decision-making scenarios involving environmental ambiguities and tactical path planning. Additionally, our findings suggest significant potential in developing model-specific prompt engineering methodologies that leverage the distinct architectural characteristics of each language model, rather than employing uniform prompting strategies across heterogeneous systems. Preliminary results indicate that tailored prompt design could substantially mitigate the performance variability observed between different model architectures. Furthermore, we envision implementing a modular visual-reasoning pipeline that decouples perceptual processing from navigational decision-making, wherein specialized visual models extract and interpret environmental features from sensory inputs before transmitting structured representations to dedicated reasoning models for trajectory planning and obstacle avoidance. This architectural separation may resolve the cross-modal integration challenges identified in current systems. Perhaps most promising is the integration of retrieval-augmented generation (RAG) mechanisms for spatial memory persistence, enabling the construction of dynamically updated environmental representations across sequential decision cycles. By systematically archiving point cloud data and visual observations within an accessible memory structure, future iterations could maintain a continuously evolving overhead mapping system, providing crucial contextual awareness for complex navigation scenarios and potentially addressing the recursive behavior patterns observed in extended operational sequences. This spatiotemporal

memory framework would effectively transform discrete perceptual inputs into a cohesive environmental model capable of informing more sophisticated navigational strategies.

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